

TG621K GNSS Module Data Sheet

Model No.: TG621K
Revision 1.1

Features

- Support BD/GPS/GLONASS/GALILEO/QZSS/SBAS multi-constellation positioning, D-GNSS, A-GNSS, Ephemeris prediction and DR(Dead Reckoning)
- 16 million time-frequency hypothesis testing per sec
- -148dBm cold start sensitivity
- -165dBm tracking sensitivity
- 27.5 second cold start TTFF
- <10 second TTFF with AGPS
- 1 second hot start
- 2.5m CEP accuracy
- Multipath detection and suppression
- Jamming detection and mitigation
- Self-Aided Ephemeris Estimation
- Contains LNA, SAW Filter, TCXO, RTC Xtal, Regulator
- Works with active and passive antenna
- Complete receiver in 12.2mm x 16.0mm size
- Operating temperature -40 ~ +85°C
- Pb-free RoHS compliant

Applications

- Navigation and asset tracking
- Timing reference
- LBS(Location Based Service)
- PND(Portable Navigation Device)
- Mobile phone

TG621K

High-performance GPS/Glonass/BDS/ GALILEO Solution Module

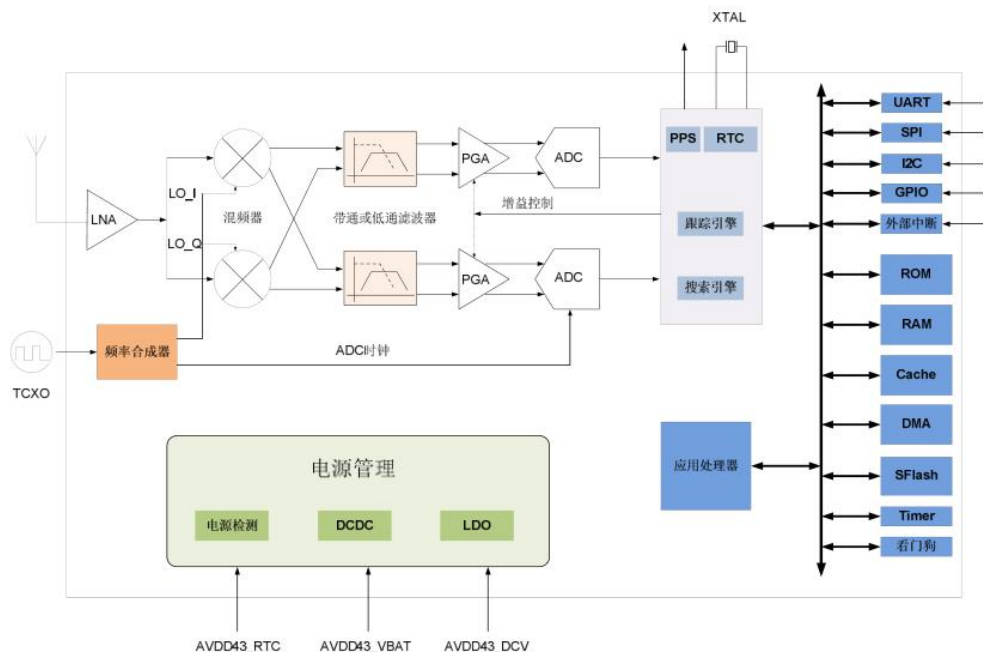
The TG621K is a high-performance GNSS all-in-one solution module that features super sensitivity, ultra low power and small form factor. The RF signal is applied to the antenna input of module, and a complete serial data message with position, velocity and time information is presented at the serial interface with NMEA protocol or custom protocol.

It is based on the high performance features of the GK9501 single-chip architecture, Its -165dBm tracking sensitivity extends positioning coverage into place like urban canyons and dense foliage environment where the GNSS was not possible before. The small form factor and low power consumption make the module easy to integrate into portable device like PNDs, mobile phones, cameras and vehicle navigation systems.

The TG621K module contains LNA for easy integration with passive antenna and a SAW filter for increased jamming immunity. It works with both passive and active antenna.

TECHNICAL SPECIFICATIONS		
TTFF	Cold start	27.5s
	Warm start	<1s
	Re-Acquisition	<1s
Sensitivity	cold-start	-148dBm
	re-acquisition	-160dBm
	tracking	-165dBm
Accuracy	Horizontal position accuracy	3m
	Altitude position accuracy	4m
	Velocity accuracy	0.1m/s
	Accuracy of timepulse signal	30ns
Power Consumption	Acquisition	33mA
Input Voltage		3V~3.6V
A-GPS		Self-aided ephemeris estimation
Serial Interface		3.3V LVTTTL level
Protocol		NMEA-0183 V3.01, 9600 baud, 8, N, 1
Dimension		16.0mm L x 12.2mm W
Operating Temperature		-40°C ~ +85°C
Storage Temperature		-55 ~ +100°C
Humidity		5% ~ 95%

BLOCK DIAGRAM



Pin Assignment

13	GND	GND	12
14	NC	RF_IN	11
15	NC	GND	10
16	NC	VCCRF	9
17	NC	NC	8
18	NC	NC	7
19	NC	NC	6
20	TXD	NC	5
21	RXD	NC	4
22	VBKP	1PPS	3
23	VCC	NC	2
24	GND	NC	1

Pin No	Pin name	I/O	Description	Remark
1	NC			
2	NC			
3	1PPS	O	Time Pulse Signal	Leave open if not used
4	NC			
5	NC			
6	NC			
7	NC			
8	NC			
9	VCCRF	O	VCC power output	
10	GND	G	Ground	
11	RF_IN	I	RF Signal Input	50Ω
12	GND	G	Ground	
13	GND	G	Ground	
14	NC			
15	NC			
16	NC			
17	NC			
18	NC			

19	NC			
20	TXD	O	UART Serial Data Output	Leave open if not used
21	RXD	I	UART Serial Data Input	Leave open if not used
22	VBKP	I	RTC and backup SRAM power	This pin may be connect to Battery or Power Supply
23	VCC	I	Module Power Supply	Operating range: 3.0V~3.6V
24	GND	G	Ground	

Interfaces Configuration

Power Supply

Regulated power for the TG621K is required. The input voltage Vcc should be 3.0V to 4.2V range, current is no less than 100mA. Suitable decoupling must be provided by external decoupling circuitry (10uF and 1uF). It can reduce the Noise from power supply and increase power stability. Main power supply Vcc current varies according to the processor load and satellite acquisition. Maximum Vcc peak current is about 45 mA during acquisition.

Backup Battery Power

In case of a power failure on pin Vcc, real-time clock and backup RAM are supplied through pin V_BCKP. This enables the TG621K Receiver to recover from power failure with either a hot start or a warm start. If no Backup Battery is connected, the receiver performs a cold start upon powered up. Backup Battery Power V_BCKP draws typically 15 uA current in backup state.

Antenna

The TG621K receiver is designed for supporting the active antenna or passive antenna connected with pin RF_IN. The gain of active antenna should be no more than 25dB (18~20dB Typical). The maximum noise figure should be no more than 1.5dB and output impedance is at 50 Ohm.

NOTE With passive antenna keep the cable loss at minimum(<1dB).

Vcc_out

Antenna power output pin. When user wants to use external active antenna. The pin supply power for active antenna.

UART Ports

The module supports one full duplex serial channels UART. The serial connections are at 2.85V LVTTTL logic levels, if need different voltage levels, use appropriate level shifters. the data format is however fixed: X, N, 8, 1, i.e. X baud rate, no parity, eight data bits and one stop bit, no other data formats are supported, LSB is sent first. The modules default baud rate is set up 9600bps. The RXD & TXD recommended to pull up (10KΩ). It can increase the stability of serial data.

RF_IN

The transmission line must to be control impedance from RF_IN pin to the antenna or antenna connector of your choice. (Impedance 50Ω)

PPS

A pulse per second (1 PPS) is an electrical signal that very precisely indicates the start of a second. Depending on the source, properly operating PPS signals have an accuracy ranging 10ns. The PPS signals are used for precise timekeeping and time measurement.

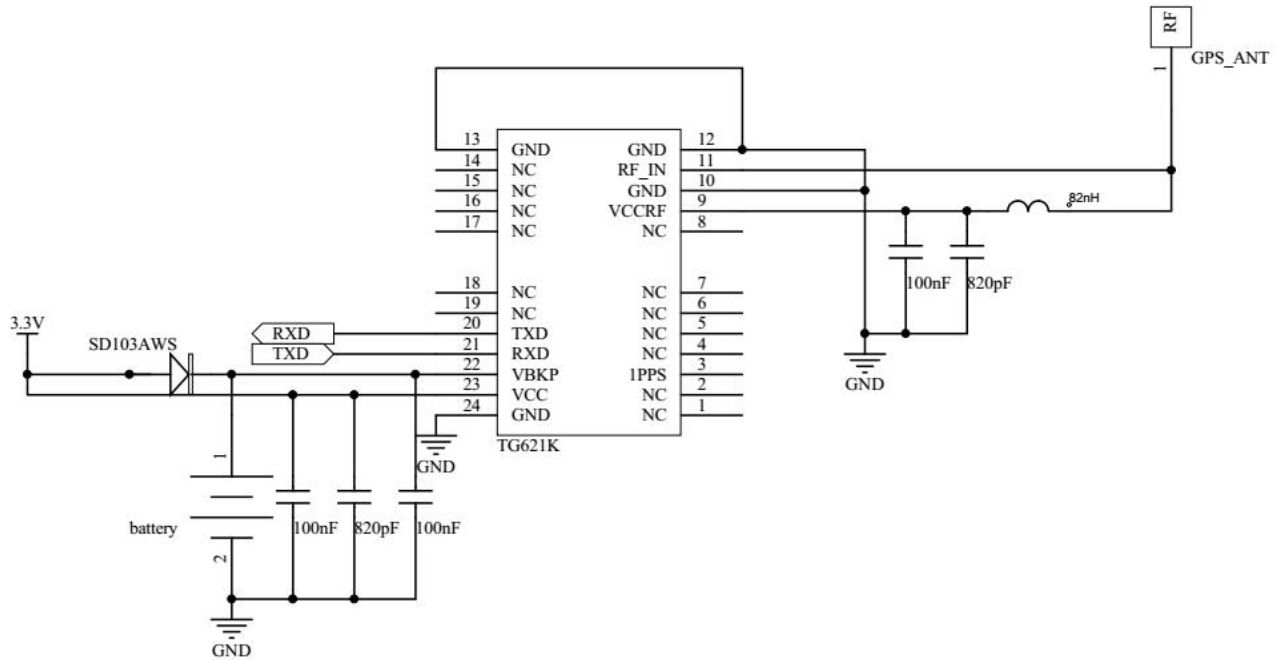
ELECTRICAL SPECIFICATIONS**ABSOLUTE MAXIMUM RATINGS**

Parameter	Minimum	Maximum	Condition
Supply Voltage (VCC)	-0.5	3.6	Volt
Backup Battery Voltage (V_BCKP)	-0.5	3.6	Volt
Input Pin Voltage	-0.5	VCC+0.5	Volt
Input Power at RF_IN		+5	dBm
Storage Temperature	-55	+100	degC

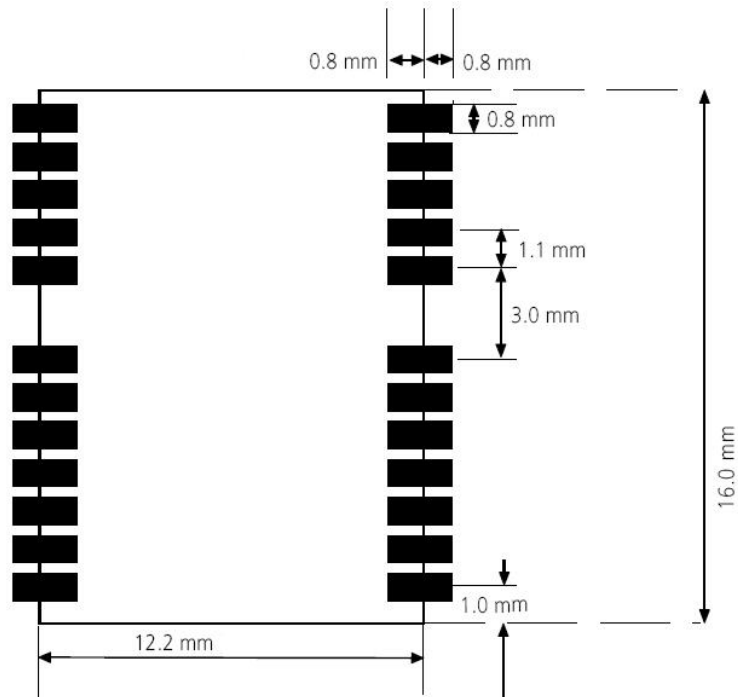
OPERATING CONDITIONS

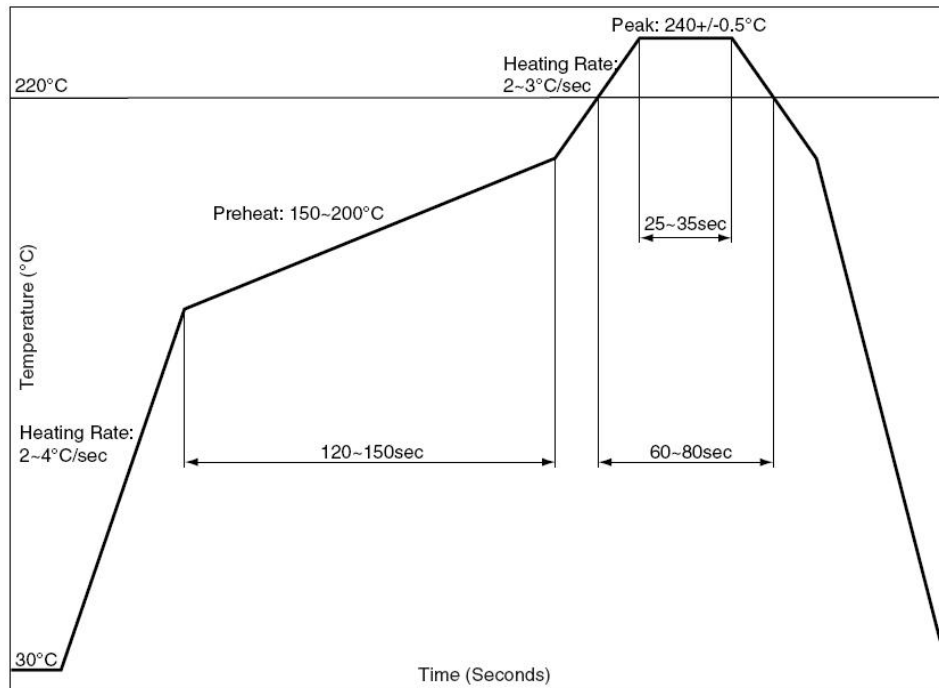
Parameter	Min	Typ	Max	Unit
Supply Voltage (VCC)	3	3.3	3.6	Volt
Acquisition Current (exclude active antenna current)		45		mA
Tracking Current (exclude active antenna current)		39		mA
Backup Voltage (V_BCKP)	2.5		3.6	Volt
Backup Current (VCC voltage applied)			0.5	mA
Backup Current (VCC voltage off)			35	uA
Output Low Voltage			0.4	Volt
Output HIGH Voltage	2.4			Volt
Input LOW Voltage			0.8	Volt
Input HIGH Voltage	2			Volt
Input LOW Current	-10		10	uA
Input HIGH Current	-10		10	uA
RF Input Impedance (RF_IN)		50		Ohm

APPLICATION CIRCUIT



RECOMMENDED FOOTPRINT



RECOMMENDED REFLOW PROFILE

The reflow profile shown above should not be exceeded, since excessive temperatures or transport times during reflow can damage the module. Cooling temperature fall rate: max 3°C / sec

ANTENNA CONSIDERATIONS

The TG621K is designed to use with a wide variety of active and passive antennas, but care must be taken during antenna selection to ensure optimum signal reception performance. There are many choices of antenna configurations; the best choice is often a tradeoff between size, gain, bandwidth and cost. The best way is to test multiple antenna solutions in the configuration of the final system to determine which provides the best overall performance.

Ceramic patch antenna is low-cost and provides good sensitivity. 50-ohm output ceramic patch antenna can be connected directly to RF input of the module. Usually the antenna and TG621K are mounted on opposite side of the PCB to reduce possibility of picking up digital noise. To improve signal reception performance, use larger ground plane under the patch antenna If possible; larger the ground plane, larger the overall antenna gain. The center frequency of the ceramic patch antenna changes with ground plane size. For optimal GPS/Beidou or GPS operation, frequency bandwidth of the antenna needs to cover 1559MHz ~ 1577MHz or 1574MHz ~ 1577MHz respectively when mounted on the PCB. It is usual to ask the ceramic patch antenna vendor to select or tune a patch antenna that best matches the customer PCB.

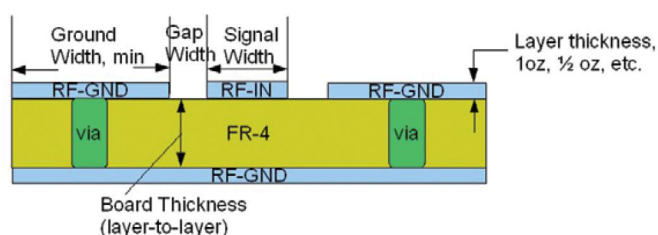
Active antenna is essentially a passive antenna with built-in LNA and a coaxial cable to connect the antenna to the module. It has the flexibility of being located remotely from the module, but requires antenna power. Active antenna usually costs more than passive patch antenna, but the performance in low signal environments is usually better. When using active antenna, an external bias choke inductor is used to provide DC bias for the active antenna. Active antenna with gain up to 30dB and noise figure less than 1.5dB can be used with TG621K.

Chip antenna is often desired for its small size. Matching element of the chip antenna needs to be designed-in according to the chip antenna datasheet. If application doesn't have a large ground plane as shown in the datasheet for the measured performance spec condition, testing will be needed to determine if it can provide acceptable performance with the smaller sized application PCB.

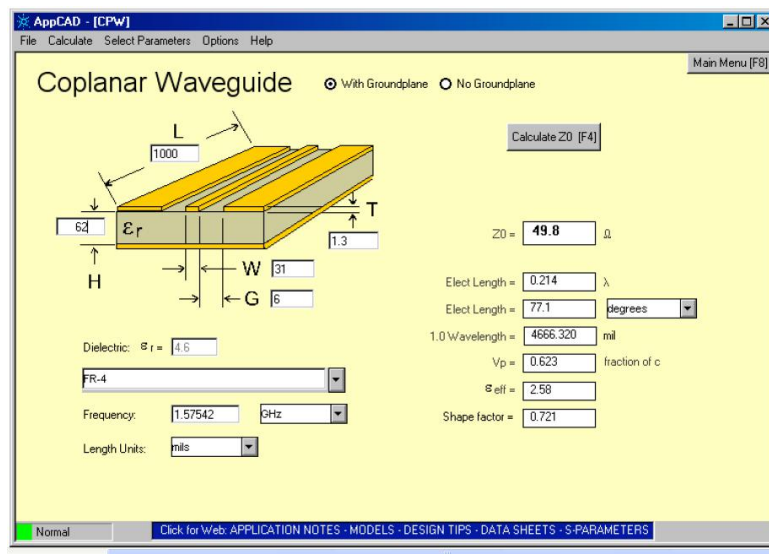
	TG621K		TG621K-GL	
Antenna Type	Passive	Active	Passive	Active
GPS Frequency (MHz)	1575.42 +/- 2	1575.42 +/- 2	1575.42 +/- 2	1575.42 +/- 2
Beidou Frequency (MHz)				

GLONASS Frequency (MHz)			1602 +/- 4	1602 +/- 4
VSWR	< 2 (typical)	< 2 (typical)	< 2 (typical)	< 2 (typical)
Polarization	RHCP or Linear	RHCP or Linear	RHCP or Linear	RHCP or Linear
Antenna Gain	> 0dBi	> -2dBi	> 0dBi	> -2dBi
LNA Gain		20dB (typical)		20dB (typical)
Noise Figure		< 1.5dB		< 1.5dB
Total Gain		> 18dBi		> 18dBi

The signal path from antenna to RF input of TG621K is the most critical part of application design. The goal is to provide optimal 50-ohm match between a 50Ω antenna and the module 50-ohm RF input for maximum power transfer. The 50-ohm grounded coplanar wave guide, consisting of the RF input signal with RF ground on either sides and a RF ground underneath, is a good choice for efficiency.



For a two-layer FR4 PCB design with 1.6mm thickness, 4.6 dielectric constant, and 1oz copper the RF-input trace should be 31mil in width, the gap to the adjacent grounds should be 6mil, and each of the RF grounds should be at least twice the width of the input signal trace (62mil). Freeware program such as AppCAD can be used to calculate values required for other configurations.



LAYOUT GUIDELINES

Separate RF and digital circuits into different PCB regions.

It is necessary to maintain 50-ohm impedance throughout the entire RF signal path. Try keeping the RF signal path as short as possible.

Do not route the RF signal line near noisy sources such as digital signals, oscillators, switching power supplies, or other RF transmitting circuit. Do not route the RF signal under or over any other components (including TG621K), or other signal traces. Do not route the RF signal path on an inner layer of a multi-layer PCB to minimize signal loss.

Avoid sharp bends for RF signal path. Make two 45-deg bends or a circular bend instead of a single 90-degree bend if needed.

Avoid vias with RF signal path whenever possible. Every via adds inductive impedance. Vias are acceptable for connecting the RF grounds between different layers. Each of the module's ground pins should have short trace tying immediately to the ground plane below through a via.

The bypass capacitors should be low ESR ceramic types and located directly adjacent to the pin they are for.

HANDLING GUIDELINE

The TG621K modules are rated MSL4, must be used for SMT reflow mounting within 72 hours after taken out from the vacuumed ESD-protective moisture barrier bag in factory condition < 30degC / 60% RH. If this floor life time is exceeded, or if the received ESD-protective moisture barrier bag is not in vacuumed state, then the device need to be pre-baked before SMT reflow process. Baking is to be done at 85degC for 8 to 12 hours. Once baked, floor life counting begins from 0, and has 72 hours of floor life at factory condition < 30degC / 60% RH.

TG621K module is ESD sensitive device and should be handled with care.

NMEA Output Description

The output protocol supports NMEA-0183 standard. The implemented messages include GGA, GLL, GSA, GSV, VTG, RMC messages. The NMEA message output has the following sentence structure:

\$aacc,c-c*hh<CR><LF>

The detail of the sentence structure is explained in Table 1.

Table 1: The NMEA sentence structure

character	HEX	Description
"\$"	24	<u>Start of sentence.</u>
Aacc		<u>Address field.</u> "aa" is the talker identifier. "cc" identifies the sentence type.
","	2C	<u>Field delimiter.</u>
C-c		<u>Data sentence block.</u>
"*"	2A	<u>Checksum delimiter.</u>
Hh		<u>Checksum field.</u>
<CR><LF>	0D0A	<u>Ending of sentence.</u> (carriage return, line feed)

Table 2: Overview of ZBTCOM receiver's NMEA messages for TG621K

\$GPGGA	Time, position, and fix related data of the receiver.
\$GPGSA	Used to represent the ID's of satellites which are used for position fix.
\$GPGSV	Satellite information about elevation, azimuth and CNR
\$GPRMC	Time, date, position, course and speed data.
\$GPVTG	Course and speed relative to the ground.

Table 3: Overview of ZBTCOM receiver's NMEA messages for TG621K-BD

\$GNGGA	Time, position, and fix related data of the receiver.
\$GPGSA \$BDGSA	Used to represent the ID's of satellites which are used for position fix.
\$GPGSV \$BDGSV	Satellite information about elevation, azimuth and CNR, \$GPGSV is used for GPS satellites, while \$BDGSV is used for Beidou satellites
\$GNRMC	Time, date, position, course and speed data.
\$GNVTG	Course and speed relative to the ground.

Table 4: Overview of ZBTCOM receiver's NMEA messages for TG621K-GL

\$GNGGA	Time, position, and fix related data of the receiver.
\$GPGSA \$GLGSA	Used to represent the ID's of satellites which are used for position fix.
\$GPGSV \$GLGSV	Satellite information about elevation, azimuth and CNR, \$GPGSV is used for GPS satellites, while \$GLGSV is used for GLONASS satellites
\$GNRMC	Time, date, position, course and speed data.
\$GNVTG	Course and speed relative to the ground.

The formats of the supported NMEA messages are described as follows:

GGA – Global Positioning System Fix Data

Time, position and fix related data for a GPS receiver.

Format:

\$--GGA,hhmmss.sss,llll.llll,a,yyyyy.yyyy,a,x,uu,v.v,w,w,M,x.x,M,,zzzz*hh<CR><LF>

Field	Name	Description
hhmmss.sss	UTC Time	UTC of position in hhmmss.sss format, (000000.000 ~ 235959.999)
llll.llll	Latitude	Latitude in ddmm.mmmm format. Leading zeros are inserted.
A	N/S Indicator	'N' = North, 'S' = South
yyyyy.yyyy	Longitude	Longitude in dddmm.mmmm format. Leading zeros are inserted.
A	E/W Indicator	'E' = East, 'W' = West
x	GPS quality indicator	GPS quality indicator 0: position fix unavailable 1: valid position fix, SPS mode 2: valid position fix, differential GPS mode
uu	Satellites Used	Number of satellites in use, (00 ~ 24)
v.v	HDOP	Horizontal dilution of precision, (0.0 ~ 99.9)
w.w	Altitude	Mean sea level altitude (-9999.9 ~ 17999.9) in meter
x.x	Geoidal Separation	In meter
zzzz	DGPS Station ID	Differential reference station ID, 0000 ~ 1023 NULL when DGPS not used
hh	Checksum	

GLL – Geographic Position – Latitude/Longitude

Latitude and longitude of vessel position, time of position fix and status.

Format:

\$--GLL,IIII.IIIII,a,yyyyy.yyyy,b,hhmmss.sss,A,a*hh<CR><LF>

Field	Name	Description
IIII.IIIII	Latitude	Latitude in ddmm.mmmm format. Leading zeros are inserted.
A	N/S Indicator	'N' = North, 'S' = South
yyyyy.yyyy	Longitude	Longitude in dddmm.mmmm format. Leading zeros are inserted.
B	E/W Indicator	'E' = East, 'W' = West
hhmmss.sss	UTC Time	UTC of position in hhmmss.sss format, (000000.000 ~ 235959.999)
A	Status	A= data valid, V= Data not valid
hh	Checksum	

GSA – GNSS DOP and Active Satellites

GPS receiver operating mode, satellites used in the navigation solution reported by the GGA or GNS sentence and DOP values.

Format:

\$--GSA,a,x,xx,xx,xx,xx,xx,xx,xx,xx,xx,xx,xx,u.u,v.v,z.z*hh<CR><LF>

Field	Name	Description
a	Mode	Mode 'M' = Manual, forced to operate in 2D or 3D mode 'A' = Automatic, allowed to automatically switch 2D/3D
x	Mode	Fix type 1 = Fix not available 2 = 2D 3 = 3D
xx's	Satellite ID	01 ~ 32 are for GPS; 33 ~ 64 are for WAAS (PRN minus 87); 65 ~ 96 are for GLONASS (64 plus slot numbers); 193 ~ 197 are for QZSS; 01 ~ 37 are for Beidou (BD PRN). GPS and Beidou satellites are differentiated by the GP and BD prefix. Maximally 12 satellites are included in each GSA sentence.
u.u	PDOP	Position dilution of precision (0.0 to 99.9)
v.v	HDOP	Horizontal dilution of precision (0.0 to 99.9)
z.z	VDOP	Vertical dilution of precision (0.0 to 99.9)
hh	Checksum	

GSV – GNSS Satellites in View

Number of satellites (SV) in view, satellite ID numbers, elevation, azimuth, and SNR value. Four satellites maximum per transmission.

Format:

\$--GSV,x,u,xx,uu,vv,zzz,ss,uu,vv,zzz,ss,...,uu,vv,zzz,ss*hh<CR><LF>

Field	Name	Description
x	Number of message	Total number of GSV messages to be transmitted (1-4)
u	Sequence number	Sequence number of current GSV message
xx	Satellites in view	Total number of satellites in view (00 ~ 16)
uu	Satellite ID	01 ~ 32 are for GPS; 33 ~ 64 are for WAAS (PRN minus 87); 65 ~ 96 are for GLONASS (64 plus slot numbers); 193 ~ 197 are for QZSS; 01 ~ 37 are for Beidou (BD PRN). GPS and Beidou satellites are differentiated by the GP and BD prefix. Maximally 4 satellites are included in each GSV sentence.
vv	Elevation	Satellite elevation in degrees, (00 ~ 90)
zzz	Azimuth	Satellite azimuth angle in degrees, (000 ~ 359)
ss	SNR	C/No in dB (00 ~ 99) Null when not tracking
hh	Checksum	

RMC – Recommended Minimum Specific GNSS Data

Time, date, position, course and speed data provided by a GNSS navigation receiver.

Format:

\$--RMC,hhmmss.sss,x,IIII.IIII,a,yyyyy.yyyy,a,x.x,u.u,xxxxxx,,v*hh<CR><LF>

Field	Name	Description
hhmmss.sss	UTC time	UTC time in hhmmss.sss format (000000.000 ~ 235959.999)
x	Status	Status 'V' = Navigation receiver warning 'A' = Data Valid
IIII.IIII	Latitude	Latitude in dddmm.mmmm format. Leading zeros are inserted.
A	N/S indicator	'N' = North; 'S' = South
yyyyy.yyyy	Longitude	Longitude in dddmm.mmmm format. Leading zeros are inserted.
A	E/W Indicator	'E' = East; 'W' = West
x.x	Speed over ground	Speed over ground in knots (000.0 ~ 999.9)
u.u	Course over ground	Course over ground in degrees (000.0 ~ 359.9)
xxxxxx	UTC Date	UTC date of position fix, ddmmyy format
v	Mode indicator	Mode indicator 'N' = Data not valid

		'A' = Autonomous mode 'D' = Differential mode 'E' = Estimated (dead reckoning) mode
hh	checksum	

VTG – Course Over Ground and Ground Speed

The actual course and speed relative to the ground.

Format:

\$--VTG,x.x,T,y.y,M,u.u,N,v.v,K,m*hh<CR><LF>

Field	Name	Description
x.x	Course	Course over ground, degrees True (000.0 ~ 359.9)
y.y	Course	Course over ground, degrees Magnetic (000.0 ~ 359.9)
u.u	Speed	Speed over ground in knots (000.0 ~ 999.9)
v.v	Speed	Speed over ground in kilometers per hour (0000.0 ~ 1800.0)
m	Mode	Mode indicator 'N' = not valid 'A' = Autonomous mode 'D' = Differential mode 'E' = Estimated (dead reckoning) mode
hh	Checksum	

ZDA – Time and Date

UTC, day, month, year and local time zone.

Format:

\$--ZDA,hhmmss.sss,dd,mm,yyyy,xx,yy*hh<CR><LF>

Field	Name	Description
hhmmss.sss	UTC time	UTC time in hhmmss.sss format (000000.000 ~ 235959.999)
dd	UTC day	01 to 31
mm	UTC month	01 to 12
yyyy	UTC year	Four-digit year number
xx	Local zone hours	00 to +-13
yy	Local zone minutes	00 to +59
hh	Checksum	

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